DM4ML – ASSIGNMENT 2

End-to-End Data Management Pipeline for an E-commerce Recommendation System

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GitHub Repository:
https://github.com/2025ag05144-MadhuRani/DM4ML-Assignment-2-Recommendation-System

1. Objective

This assignment implements an end-to-end data management pipeline for an e-commerce recommendation system. The stages include synthetic data ingestion, data validation, preparation, feature engineering, model training, evaluation, and workflow verification.

2. Dataset and Ingestion

Three synthetic CSV datasets were generated and stored in data/raw/.

Dataset	Rows	Columns	Description
users.csv	500	4	User profile information
items.csv	200	4	Product/item information
interactions.csv	12,000	6	User-item interaction records

3. Validation and Preparation

Validation covered missing values, duplicate records and interaction IDs, data types, rating validity, user/item referential integrity, timestamps, and item prices. All 12 recorded validation checks passed; the generated datasets had zero missing values, duplicates, and invalid records. The report was saved as reports/data_validation_report.csv.

Cleaned datasets were saved in data/processed/. Row counts remained 500 users, 200 items, and 12,000 interactions. Interactions were sorted chronologically.

4. Feature Engineering

Feature group	Engineered features	Output shape
User	interaction_count, user_avg_rating, preferred_category	500 × 7
Item	interaction_count, item_avg_rating, popularity_score	200 × 7
Interaction	days_since_interaction, category	12,000 × 8

The feature catalog contains eight engineered features across user, item, and interaction groups and is saved as reports/feature_catalog.csv.

5. Chronological Split and Model Training

The interaction data was split chronologically using an 80:20 ratio to avoid training on future interactions. The training set contains 9,600 records and the test set contains 2,400 records.

Split	Rows	Time range
Training	9,600	2025-01-01 to 2025-10-20 01:00
Testing	2,400	2025-10-20 01:00 to 2025-12-31 00:00

Popularity baseline: ranks items using interaction frequency. Output: models/popularity_model.csv.

Personalized item-based collaborative filtering: creates a user-item matrix, computes item-item cosine similarity, and generates personalized recommendations. The similarity matrix and sample recommendations are stored under models/.

6. Model Evaluation

Evaluation used test users and treated ratings of 4 or higher as relevant. Ranking metrics were Precision@10, Recall@10, NDCG@10, Coverage, and Diversity.

Metric	Popularity baseline	Item-based CF
Precision@10	0.012297	0.010441
Recall@10	0.048884	0.045343
NDCG@10	0.024728	0.026468
Coverage	0.085	1.000
Diversity	0.856458	0.809126

Popularity achieved slightly higher Precision@10 and Recall@10. Item-based CF achieved higher NDCG@10 and full item coverage. Popularity achieved higher diversity. These results use synthetic data and demonstrate the evaluation process; they do not establish production recommendation quality.

Results were saved in reports/model_evaluation_results.csv. Comparison charts are reports/precision_recall.png, reports/ndcg_comparison.png, and reports/coverage_diversity.png.

7. Workflow Verification

The script dags/recommendation_pipeline.py checks whether expected output files exist for the six pipeline stages and writes statuses and timestamps to reports/workflow_execution_log.csv.

Workflow stage	Execution status
Data Ingestion	SUCCESS
Data Validation	SUCCESS
Data Preparation	SUCCESS
Feature Engineering	SUCCESS
Model Training	SUCCESS
Model Evaluation	SUCCESS

Implementation note: this is a standalone Python artifact-verification workflow, not an Apache Airflow DAG or scheduler. It verifies outputs from stages already executed; it does not rerun the pipeline.

8. Project Structure and Reproducibility

data/raw/ – generated source datasets
data/processed/ – cleaned datasets and chronological train/test splits
data/features/ – engineered feature datasets
data/errors/ – reserved for quarantined invalid records
src/ – modular ingestion, validation, preparation, transformation, features, training, and evaluation areas
dags/ – workflow verification script
models/ – model outputs and recommendations
reports/ – validation, feature catalog, evaluation, charts, and workflow log
configs/ and experiments/ – configuration and experiment organization

The root requirements.txt lists pandas, NumPy, scikit-learn, matplotlib, seaborn, MLflow, and ReportLab. The README documents the project objective, datasets, pipeline stages, models, and evaluation summary.

9. Conclusion

The assignment demonstrates an end-to-end recommendation-system data pipeline using synthetic e-commerce data. It includes ingestion, validation, chronological preparation, feature engineering, popularity-based and personalized recommendation models, ranking evaluation, and workflow verification. All six workflow artifact checks completed with SUCCESS status.

Item-based collaborative filtering achieved full item coverage and higher NDCG@10 in this experiment, while the popularity baseline performed slightly better on Precision@10, Recall@10, and Diversity. The results illustrate trade-offs between approaches and provide a foundation for further work using real interaction data and a production workflow orchestrator.

Deliverables: GitHub repository and README; raw, processed, and feature datasets; validation report; feature catalog; model outputs; evaluation results and charts; workflow verification script and execution log; final report.

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